

18.06 Problem Set 13

Due **Wednesday, December 7** at 11am. This is the **last pset in 18.06 this term** and the **last pset covered on exam 3**.

Problem 1 (5 + 5 points)

For the following parts, use one (or more) of the equivalent properties (from class) that define positive-definiteness. (There are multiple possible approaches.)

(a) If A and B are positive-definite $m \times m$ matrices, why must their sum $A + B$ be positive definite?

(b) If B is a positive-definite $m \times m$ matrix and C is an $m \times n$ matrix with full column rank, why must $C^H B C$ be positive-definite?

Problem 2 (5 points)

In class, we analyzed a system with n masses and $n + 1$ springs and showed that it satisfied an equation

$$m \frac{d^2 x}{dt^2} = -D^T K D x$$

for the vector $x \in \mathbb{R}^n$ of displacements, and we showed that $-D^T (K/m) D$ was negative-definite, and that this led to oscillating solutions.

Suppose that the masses are *not* identical, and let M be the diagonal $n \times n$ matrix of masses $m_1, \dots, m_n > 0$. If we define $y = \sqrt{M} x$, then show that $\frac{d^2 y}{dt^2} = B y$ where B is negative-definite, and hence we *still* have oscillating solutions.

Problem 3 (5 + 5 + 5 points)

The nullspace $N(A)$ of the real matrix A is spanned by the vector $v = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

(a) Give an eigenvector and eigenvalue of the matrix $B = (3I - A^T A)(3I + A^T A)^{-1}$.

(b) Aside from the eigenvalue identified in (a), if you consider all *other* eigenvalues λ of B , which of the following *must* be true? (Indicate *all* that apply.)

- purely real, purely imaginary, zero, negative real part, positive real part, $|\lambda| < 1$, $|\lambda| > 1$, $|\lambda| \leq 1$, and/or $|\lambda| \geq 1$?

Justify your answer.

(c) Give a good approximate formula for $B^n \begin{pmatrix} 0 \\ -1 \\ 0 \\ 8 \end{pmatrix}$ for a large n (give an explicit numerical vector).